Lecture 13: Solving Non-Linear Equations

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Root Finding – Financial Example

Often, we need to find the zeros/roots of a non-linear equation.

Consider the no-arbitrage price of an option given as:

$P = f(S, T, \sigma, r)$, where S = current price, T = expiry, σ = volatility, r = interest rate.

Here, f is a function (or program) that gives the price, e.g., a lattice price or a function, like `blspice`, that returns the Black-Scholes solution.

Implied volatility problem: Given the observed market price P_{obs}, and the values (T, S, r), what is the volatility σ that this implies?

Root Finding – Financial Example

Mathematically, then, our task is to find $\sigma = \sigma_{imp}$ such that:

$$P_{obs} - f(S, T, \sigma, r) = 0$$

i.e., what volatility σ makes the *observed* price P_{obs} consistent with our mathematical model f for the price?

This is a non-linear equation for σ_{imp}. We must find its roots / zeros.

Why do we need numerical methods for root finding?

Question: Why do we need numerical methods for root finding?

Suppose $N(t)$ denotes the number in the population at time t, and λ denotes the constant birth rate of the population.

Then the population satisfies the differential equation $\frac{dN}{dt} = \lambda N(t)$

with the solution $N(t) = N_0 e^{\lambda t}$, where N_0 is the initial population.

Now if *immigration* is permitted at a constant rate ν, then:

$\frac{dN}{dt} = \lambda N(t) + \nu$, with $N(t) = N_0 e^{\lambda t} + \frac{\nu}{\lambda}(e^{\lambda t} - 1)$ as its solution.

Why do we need numerical methods for root finding?

After one year:

$$N(1) = N_0 e^{\lambda(1)} + \frac{\nu}{\lambda}\left(e^{\lambda(1)} - 1\right)$$

To find λ, we must solve this equation for λ. However, it is impossible to solve explicitly for λ in this equation!

However, numerical methods can approximate solutions of equations of this type with arbitrarily high accuracy.

General Root-Finding Problem

Problem: Given any function $f(x)$, find x^* such that $f(x^*) = 0$.
x^* is called a **root** of the equation $f(x) = 0$.

$\sqrt{2}$

$x^2 - 4 = 0$

Challenges:

- Even if the existence of a root of a function is guaranteed, there is no guaranteed method for finding x^*
- Since x^* may not be defined in our floating point system, we cannot find x^* exactly.

What should we do?

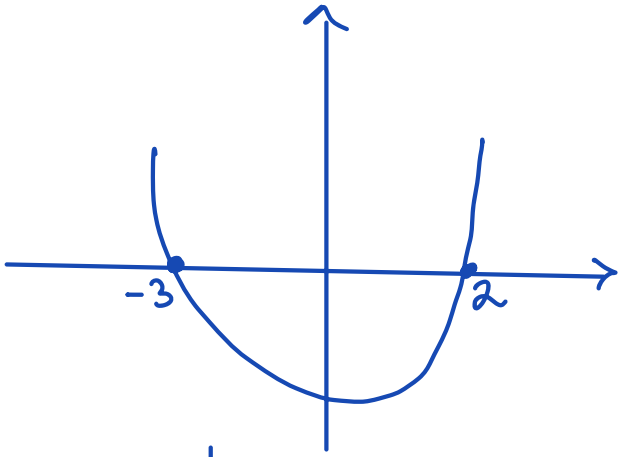
Computational Solution: Given any $f(x)$ and some error tolerance $\epsilon > 0$, find x^* such that $|f(x^*)| < \epsilon$.

Examples

Ex1: $f(x) = x^2 + x - 6$

$= (x-2)(x+3)$

has two roots: $x^* = 2, -3$

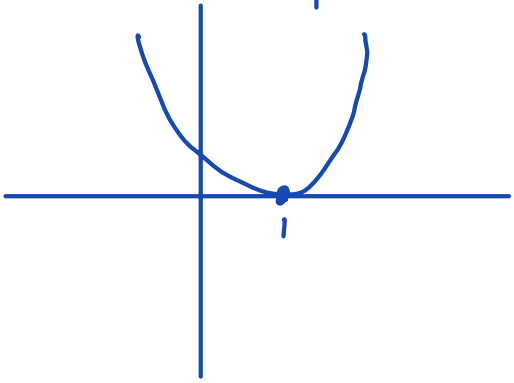


Ex2: $f(x) = x^2 - 2x + 1$

$= (x-1)^2$

has one double root,

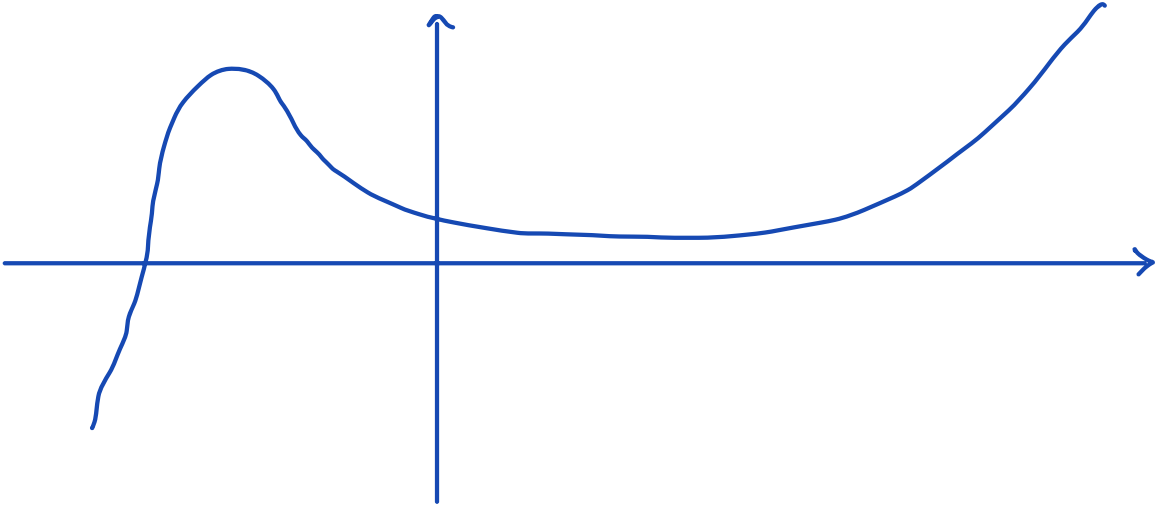
$x^* = 1$



Definition: We say x^* is a ***double root*** of $f(x) = 0$ *iff* $f(x^*) = 0$ and $f'(x^*) = 0$.

Examples

Ex3: $f(x) = 3x^5 + 5x^4 + \frac{1}{3}x^3 + 1$



Examples

Ex4: $f(x) = x^2 - \frac{1}{2}e^{-x}$

$x=0,\ f(0) = \frac{-1}{2} < 0$ $\longrightarrow$ $x \in [0,1]$

$x=1,\ f(1) = 1 - \frac{1}{2}e^{-1} > 0$

End of Lecture 13